

Project Report on:

**Spectral family and
Long-time quantum Dynamics.
RAGE Theorem**

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Chapter 1

Introduction

This project studies how the spectrum of a Hamiltonian operator lets us understand the long-time dynamics of the quantum system. The RAGE theorem is an important result in this area (named after David Ruelle, Walter Amrein, Vladimir Georgescu, and Volker Enß) [2][3][4]. The theorem decomposes the Hilbert space orthogonally into bound states and scattering states (states that escapes to infinity). These will correspond to the pure point subspace and the continuous subspace w.r.t. the Hamiltonian respectively.

Chapter 2 introduces the basic Mathematical framework required to understand and use the RAGE theorem. I have tried to include all the definitions and results required to understand this report. The chapter discusses Hilbert spaces, self-adjoint operators, unitary operators, Borel measures, Lebesgue measure, Sobolev spaces, and Spectral family.

Chapter 3 discusses the RAGE theorem and its applications to some class of potentials. It will be shown that free particles escapes to infinity. It will also be shown that (under some suitable assumptions) stationary states are bounded. An important result regarding a wide class of potentials will be discussed, which includes Coulomb potential.

Chapter 4 discusses scattering theory and asymptotic completeness. RAGE theorem helps to prove another important theorem (Enß) which states that short range potentials are asymptotically complete.

Majority of this report is based on ‘Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators. 2nd ed’ by Gerald Teschl [1]. I haven’t included any proofs of the results used. Most of the proofs are given in [1].

Chapter 2

Hilbert Space, Operators, Measures and Spectral Theorem

Let $\mathcal{H}$ be a vector space. A map $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ satisfying the following properties

- i) $\langle \alpha f_1 + \beta f_2, g \rangle = \alpha^* \langle f_1, g \rangle + \beta^* \langle f_2, g \rangle \quad \forall \alpha, \beta \in \mathbb{C}$
- ii) $\langle f, \alpha g_1 + \beta g_2 \rangle = \alpha \langle f, g_1 \rangle + \beta \langle f, g_2 \rangle \quad \forall \alpha, \beta \in \mathbb{C}$
- iii) $\langle f, f \rangle > 0 \quad \forall f \neq 0$
- iv) $\langle f, g \rangle = \langle g, f \rangle^* \quad \forall f, g \in \mathcal{H}$

is called an **inner product** on $\mathcal{H}$. It induces a norm $\|f\| = \sqrt{\langle f, f \rangle}$ on $\mathcal{H}$.

Hilbert Space

An inner product space $\mathcal{H}$ is called a Hilbert space if it is complete with respect to the norm induced by the inner product.

A topological space is called **separable** if it contains a countable dense subset.

Separability of Hilbert Spaces

A Hilbert space is separable $\iff$ it has a countable orthonormal basis.

Now onwards, I will use the term Hilbert space (and the symbol $\mathcal{H}$) for separable Hilbert spaces unless specified otherwise.

A linear map $A : \mathcal{D}(A) \rightarrow \mathcal{H}_2$ will be called an **operator** (or linear operator). Where $\mathcal{D}(A) \subset \mathcal{H}_1$ is a subspace and is called the domain of A . An operator $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ is called **bounded** if $\exists M > 0$ such that

$$\|Af\| \leq M \|f\| \quad \forall f \in \mathcal{D}(A).$$

Or equivalently, if $\|A\| := \sup\{\|Af\| : f \in \mathcal{D}(A), \|f\| = 1\} < \infty$.

Theorem: An operator A is bounded $\iff$ it is continuous.

Riesz Representation Theorem

Let $l : \mathcal{H} \rightarrow \mathbb{C}$ be a bounded linear functional. Then $\exists! \varphi \in \mathcal{H}$ such that $l(\psi) = \langle \varphi, \psi \rangle \quad \forall \psi \in \mathcal{H}$.

Hence, the map $\varphi \mapsto \langle \varphi, \cdot \rangle$ is an anti-linear isometric bijection between $\mathcal{H}$ and its dual space $\mathcal{H}^*$.

2.1 Self Adjoint and Unitary Operators

Adjoint of an Operator

Let $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ be an operator s.t. $\overline{\mathcal{D}(A)} = \mathcal{H}$. Then there exists a unique operator $A^\dagger : \mathcal{D}(A^\dagger) \rightarrow \mathcal{H}$ such that

$$\langle A\phi, \psi \rangle = \langle \phi, A^\dagger \psi \rangle \quad \forall \phi \in \mathcal{D}(A), \psi \in \mathcal{D}(A^\dagger).$$

Hermitian and Self-Adjoint Operators

A densely defined operator $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ is called **Hermitian** if

$$\langle A\phi, \psi \rangle = \langle \phi, A\psi \rangle \quad \forall \phi \in \mathcal{D}(A), \psi \in \mathcal{D}(A) \text{ and } \mathcal{D}(A) \subset \mathcal{D}(A^\dagger).$$

A is called **Self-Adjoint** if $\mathcal{D}(A) = \mathcal{D}(A^\dagger)$.

If $A^\dagger$ is densely defined, then we can define the closure of A as $\overline{A} := A^{\dagger\dagger}$ which is an extension of A . Further if A is Hermitian, then $\overline{A}$ is also Hermitian.

If A is Hermitian (but not self-adjoint) and $\overline{A}$ is self-adjoint, then A is called **essentially self-adjoint**.

Isometry and Unitary Operators

Let U be an operator on $\mathcal{H}$ with domain $\mathcal{D}(U)$. U is called an **isometry** if $\langle Ux, Uy \rangle = \langle x, y \rangle \quad \forall x, y \in \mathcal{D}(U)$.

Further U is called **unitary** if $U^\dagger U = UU^\dagger = \mathbb{I}$.

2.2 Borel Measures and Integration

σ -algebra

A collection of subsets $\mathcal{A}$ of a set X is called a **σ -algebra** if it satisfies:

- i) $X \in \mathcal{A}$
- ii) $\mathcal{A}$ is closed under **complements**.
- iii) $\mathcal{A}$ is closed under **countable unions**.

Borel σ -algebra and Borel Sets

If X is a topological space, then the σ -algebra generated by the open sets of X is called the **Borel σ -algebra** $\mathcal{B}(X)$ of X . Its elements are called **Borel sets**.

Borel Measure

A set X together with a σ -algebra Σ is called a **measurable space** and its elements are called **measurable sets**.

A **measure** is a map $\mu : \Sigma \rightarrow [0, \infty]$ such that:

i) $\mu(\emptyset) = 0$

ii) μ is σ -additive, i.e. if $A_j \cap A_k = \emptyset$ for all $j \neq k$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

A measure μ on the Borel σ -algebra of a topological space is called a **Borel measure** if $\mu(K) < \infty$ for every compact set K .

Let (X, Σ_X) and (Y, Σ_Y) be measurable spaces. A map $f : X \rightarrow Y$ is called **measurable** if $f^{-1}(A) \in \Sigma_X \quad \forall A \in \Sigma_Y$. If X is a topological space and Σ the corresponding Borel σ -algebra, then it is called a **Borel function**.

Integration

Let (X, Σ, μ) be a measure space. A measurable function $s : X \rightarrow \mathbb{R}$ is called **simple** if its image is finite, i.e.

$$s(x) = \sum_{i=1}^n \alpha_i \chi_{A_i}, \quad A_i = s^{-1}(\alpha_i) \in \Sigma.$$

For a non-negative simple function s , we define its integral over a set A as

$$\int_A s d\mu = \sum_{i=1}^n \alpha_i \mu(A \cap A_i).$$

The integral of a non-negative measurable function f is defined by

$$\int_A f d\mu = \sup_{s \leq f} \int_A s d\mu$$

where the supremum is taken over all simple functions pointwise less than or equal to f .

We call an arbitrary measurable function f **integrable** if the integral is finite for both the positive and negative part $f^\pm := \max\{\pm f, 0\}$. And define

$$\int_A f d\mu = \int_A f^+ d\mu - \int_A f^- d\mu.$$

Similarly for a complex-valued function, split into real and complex.

Note that f is integrable $\iff |f|$ is integrable.

2.3 Lebesgue Measure

There exists a unique measure on $\mathbb{R}^n$ that assigns to every box its usual volume and is translationally invariant. Choose this measure and also modify the σ -algebra such that a set $A \subset \mathbb{R}^n$ is said to be **Lebesgue measurable** if there exists a Borel set B such that $(A \setminus B) \cup (B \setminus A)$ has measure zero. The definitions of **Lebesgue measurable functions** and **Lebesgue integrals** follows.

Define the L^p spaces

$$L^p(X) := \{[f] : f \text{ is Lebesgue measurable, } f_1 \sim f_2 \text{ if they agree almost everywhere, } \|f\|_p < \infty\}$$

where

$$\|f\|_p := \left(\int_X |f|^p d\mu \right)^{1/p}.$$

Define the space of **locally p -integrable functions** $L^p_{\text{loc}}(X)$ as the set of all f such that $f \in L^p(K) \forall$ compact $K \subset X$.

Denote the **Lebesgue space of bounded functions vanishing at ∞** by $L^\infty(\mathbb{R}^n)$.

2.4 Sobolev Spaces

Denote the **Fourier transform** of a function $f(x)$ as $\hat{f}(p)$,

$$\hat{f}(p) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{-ipx} f(x) d^n x.$$

The **Sobolev spaces** are defined as

$$H^r(\mathbb{R}^n) := \left\{ f \in L^2(\mathbb{R}^n) \mid |p|^r \hat{f}(p) \in L^2(\mathbb{R}^n) \right\}.$$

These are Hilbert spaces with the inner product

$$\langle f, g \rangle := \int_{\mathbb{R}^n} \hat{f}(p)^* \hat{g}(p) (1 + |p|^2)^r d^n p.$$

Let $X \subset \mathbb{R}^n$. Let $C_c^\infty(X)$ denote the set of infinitely differentiable functions with compact support in X . $h \in L_{loc}^1(X)$ satisfying

$$\int_X \varphi(x) h(x) d^n x = (-1)^{|\alpha|} \int_X (\partial_\alpha \varphi)(x) f(x) d^n x \quad \forall \varphi \in C_c^\infty(X),$$

is called the **α -th weak derivative** of f . Then $H^r(\mathbb{R}^n)$ is the set of all functions having (weak) derivatives up to order r , which are in $L^2(\mathbb{R}^n)$. Then there exists an equivalent norm for this space given by $\|f\|_{2,r} = \left(\sum_{|\alpha| \leq r} \|\partial_\alpha f\|_2^2 \right)^{1/2}$.

2.5 Spectral Theorem

Denote the Borel σ -algebra on $\mathbb{R}$ by $\mathfrak{B}$. Let $\mathfrak{L}(\mathcal{H})$ be the boundary linear operators on $\mathcal{H}$.

Projection Valued Measure

A projection valued measure is a map $P : \mathfrak{B} \rightarrow \mathfrak{L}(\mathcal{H})$ such that

- i $P(\Omega)^\dagger = P(\Omega)$, $P(\Omega)^2 = P(\Omega) \quad \forall \Omega \in \mathfrak{B}$.
- ii $P(\mathbb{R}) = \mathbb{I}$.
- iii If $\Omega_n \cap \Omega_m = \emptyset$ for $n \neq m$, then $\sum_n P(\Omega_n) \psi = P(\cup_n \Omega_n) \psi \quad \forall \psi \in \mathcal{H}$.

To every projection valued measure P , there corresponds a **resolution of the identity** $P(\lambda) := P((-\infty, \lambda])$ such that

1. $P(\lambda)^\dagger = P(\lambda)$, $P(\lambda)^2 = P(\lambda)$.
2. $\lambda_1 \leq \lambda_2 \implies P(\lambda_1) \leq P(\lambda_2)$.
3. $\lim_{\lambda' \rightarrow \lambda^+} P(\lambda') \psi = P(\lambda) \psi \quad \forall \psi \in \mathcal{D}(P(\lambda))$
4. $\lim_{\lambda \rightarrow -\infty} P(\lambda) \psi = 0$, $\lim_{\lambda \rightarrow \infty} P(\lambda) \psi = \psi$

Spectral Theorem

To every self-adjoint operator A , there exists a unique projection valued measure P_A such that

$$A = \int_{\mathbb{R}} \lambda dP_A(\lambda)$$

For A self-adjoint, for each $\psi \in \mathcal{H}$, the **spectral measure** μ_ψ is defined by

$$\mu_\psi(\Omega) := \langle \psi | P_A(\Omega) \psi \rangle.$$

The **resolvent set** of an operator A is defined by

$$\rho(A) := \{z \in \mathbb{C} \mid (A - z\mathbb{I}) : \mathcal{D}(A) \rightarrow \mathcal{H} \text{ is bijective and its inverse is bounded}\}$$

The **spectrum** of A is defined by $\sigma(A) := \mathbb{C} \setminus \rho(A)$.

Theorem: Let A be symmetric. Then A is self-adjoint $\iff \sigma(A) \subset \mathbb{R}$.

Theorem: Let A be self-adjoint. Then

$$\sigma(A) = \{\lambda \in \mathbb{R} \mid P_A((\lambda - \epsilon, \lambda + \epsilon)) \neq 0 \forall \epsilon > 0\}.$$

Absolutely Continuous and Mutually Singular

A measure μ is called **absolutely continuous** w.r.t. another measure λ on the same measurable space if $\lambda(A) = 0 \implies \mu(A) = 0$.

μ and λ are called **mutually singular** if there exists a measurable set A such that $\mu(A) = 0 = \lambda(A^c)$.

Decomposition of $\mathcal{H}$

Let A be self-adjoint. $\mathcal{H}$ can be decomposed as $\mathcal{H} = \mathcal{H}_c \oplus \mathcal{H}_{pp}$

where the continuous subspace $\mathcal{H}_c := \{\psi \in \mathcal{H} \mid \mu_\psi(\{\lambda\}) = 0 \forall \lambda \in \mathbb{R}\}$

(this corresponds to continuity of $P_A(\lambda)$)

and the pure point subspace $\mathcal{H}_{pp} := \{\psi \in \mathcal{H} \mid \mu_\psi(\{\lambda\}) \neq 0 \text{ for some } \lambda \in \mathbb{R}\}$
($P_A(\lambda)$ being a step function).

It is easy to see that the pure point subspace is the closure of span of all eigenvectors of A .

$\mathcal{H}_c$ can be further divided into $\mathcal{H}_{ac} \oplus \mathcal{H}_{sc}$, where $\mathcal{H}_{ac}$ is the subspace where the spectral measure is absolutely continuous w.r.t the Lebesgue measure and $\mathcal{H}_{sc}$ is the subspace where the spectral measure is singular w.r.t. the Lebesgue measure and is continuous.

Note that $\mathcal{H}_{pp}$ is the case where the spectral measure is singular w.r.t. the Lebesgue measure and is discontinuous.

Let A be self-adjoint.

The **continuous spectrum** and **pure point spectrum** of A are defined by $\sigma_c(A) := \sigma(A|_{\mathcal{H}_c})$ and $\sigma_{pp}(A) := \sigma(A|_{\mathcal{H}_{pp}})$. Note that these may not be mutually exclusive. Similarly, we can define σ_{ac} and σ_{sc} .

The **discrete spectrum** $\sigma_d(A)$ is the set of all eigenvalues which are isolated points of the spectrum and whose corresponding eigenspace is finite dimensional.

$\sigma_{ess}(A) := \sigma(A) \setminus \sigma_d(A)$ is the **essential spectrum**. Note that the **continuous spectrum** $\sigma_c(A) \subset \sigma_{ess}(A)$ and the **pure point spectrum** $\sigma_{pp}(A) \supset \sigma_d(A)$.

Chapter 3

RAGE Theorem

An operator is called a **finite rank operator** if its range is finite dimensional, i.e. its rank is finite.

Compact Operators

The closure of the set of all finite rank operators in $\mathfrak{L}(\mathcal{H})$ is called the set of **compact operators** $\mathfrak{C}(\mathcal{H})$.

The operator K is called **relatively compact** w.r.t A if $K(A - z\mathbb{I})^{-1} \in \mathfrak{C}(\mathcal{H})$ for some $z \in \rho(A)$. (And hence $\forall z \in \rho(A)$)

3.1 RAGE Theorem

RAGE Theorem

Let A be self-adjoint. Suppose $K_n \in \mathfrak{L}(\mathcal{H})$ is a sequence of relatively compact operators which converges (for each ψ in domain) to the identity. Then

$$\mathcal{H}_c = \{\psi \in \mathcal{H} \mid \lim_{n \rightarrow \infty} \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \|K_n e^{-itA} \psi\| dt = 0\}$$

$$\mathcal{H}_{pp} = \{\psi \in \mathcal{H} \mid \lim_{n \rightarrow \infty} \sup_{t \geq 0} \|(\mathbb{I} - K_n) e^{-itA} \psi\| = 0\}$$

RAGE theorem helps to study the asymptotic behavior of quantum systems. In which case we put A to be the Hamiltonian and K_n to be operators that confine the states to a bounded region. Mostly we will be using $K_n = \chi_{B_n}$ where B_n is the ball of radius n centered at the origin and χ_B is the characteristic function of B , i.e. $\chi_B(x) = 1$ if $x \in B$ and $\chi_B(x) = 0$ if $x \notin B$.

Stationary States

Consider a system with a self-adjoint Hamiltonian H . Let $\psi_E \in \mathcal{D}(H)$ be an **energy eigenstate**. Then clearly $\psi_E \in \mathcal{H}_{pp}$. If χ_{B_n} is **relatively compact w.r.t H** $\forall n$, then by RAGE theorem,

$\forall \epsilon > 0$, $\exists n \in \mathbb{N}$ such that the probability of finding the state ψ_E outside B_n is less than ϵ . i.e. **it is a bound state**.

3.2 Free Particle

Consider a free particle in $\mathbb{R}^n$, in the Hilbert space $\mathcal{H} = L^2(\mathbb{R}^n)$. The free Hamiltonian is ($\hbar = 1, m = \frac{1}{2}$) $H_0 = -\Delta$, where $\Delta = \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$ (weak derivatives). $\mathcal{D}(H_0) = H^2(\mathbb{R}^n)$.

Theorem (7.17 in [1]): H_0 is self-adjoint and has a purely (absolutely) continuous spectrum. Hence, $\mathcal{H} = \mathcal{H}_c(H_0)$.

By Lemma 7.21 in [1], for bounded $\Omega \subset \mathbb{R}^n$, χ_Ω is relatively compact w.r.t. H_0 .

Asymptotic Behavior of a Free Particle

Hence, according to RAGE theorem, for any $\psi \in \mathcal{H}$,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left\| \chi_\Omega e^{-itH_0} \psi \right\| dt = 0$$

for any bounded $\Omega \subset \mathbb{R}^n$. This means (whatever the initial state ψ is) the particle spends negligible fraction of time in any bounded region. The time averaged probability for finding the particle in any bounded region tends to zero.

A free particle will eventually escape to infinity.

3.3 Harmonic Oscillator

The harmonic oscillator $H = H_0 + \frac{1}{2}m\omega^2 x^2 = \omega(\frac{1}{2} + \hat{N})$ is a simple case where we can see the implication of boundedness. By writing H as a countable sum over eigenspace projection operators for each eigenvalue, it will follow that the χ_Ω is relatively compact w.r.t. H and moreover $\mathcal{H} = \mathcal{H}_{pp}(H)$. Hence by RAGE theorem, **all states are bounded**.

3.4 Particle in a bounded potential

The following result (Theorem 10.2 in [1]) helps to study an important class of potentials, including Coulomb potential.

Bounded Potentials

Let V be real valued and $V \in L^\infty(\mathbb{R}^n)$ if $n > 3$ and $V \in L^\infty(\mathbb{R}^n) + L^2(\mathbb{R}^n)$ if $n \leq 3$.

Then V is relatively compact w.r.t. H_0 , and $H := H_0 + V$, $\mathcal{D}(H) = H^2(\mathbb{R}^n)$ is **self-adjoint, bounded below** and

$$\sigma_{ess}(H) = [0, \infty).$$

Further, χ_Ω is **relatively compact w.r.t. H** for any bounded $\Omega \subset \mathbb{R}^n$ and hence we can use RAGE theorem. Particularly, (physical) **energy eigenstates are bounded**.

Coulomb potential $V := \frac{\gamma}{|x|}$ for $n = 3$ and γ real satisfies the above hypothesis.

Let's look at few implications of the theorem for V satisfying the above hypothesis.

Negative energy solutions

Negative energy spectra has only isolated points and each negative energy has at most finite degeneracy. $\inf \sigma(H) > -\infty$. And of course the negative (indeed any) energy eigenstates are bounded.

Continuous spectrum states

If $\psi \in \mathcal{H}_c$, then it belongs to the positive (or 0) energy spectrum states. Particularly, states that spend negligible fraction of time in any bounded region has positive energy spectra.

Chapter 4

Scattering

In scattering, we expect the time evolution to be asymptotically free (at least for short range forces). i.e. $\|\psi(t) - \psi_{\pm}(t)\| \rightarrow 0$ as $t \rightarrow \pm\infty$. Where $\psi_{\pm}(t)$ are the asymptotic states (outgoing/incoming). This can be rewritten as

$$\lim_{t \rightarrow \pm\infty} \|\psi(0) - e^{itH} e^{-itH_0} \psi_{\pm}(0)\| = 0.$$

Hence, we define the **wave operators (Møller operators)** by

$$\Omega_{\pm} \psi := \lim_{t \rightarrow \pm\infty} e^{itH} e^{-itH_0} \psi, \quad \mathcal{D}(\Omega_{\pm}) := \{\psi \in \mathcal{H} \mid \Omega_{\pm} \psi \text{ exists}\}.$$

A state $\psi \in \text{Ran}(\Omega_+) \cap \text{Ran}(\Omega_-)$ is called a **scattering state**.

$\mathcal{D}(\Omega_{\pm})$ and $\text{Ran}(\Omega_{\pm})$ are closed and $\Omega_{\pm} : \mathcal{D}(\Omega_{\pm}) \rightarrow \text{Ran}(\Omega_{\pm})$ is **unitary**.

The **scattering operator** is defined as

$$S := \Omega_+^{-1} \Omega_-, \quad \mathcal{D}(S) := \{\psi \in \mathcal{D}(\Omega_-) \mid \Omega_- \psi \in \text{Ran}(\Omega_+)\}.$$

$S : \mathcal{D}(S) \rightarrow \text{Ran}(S)$ is **unitary** and $H_0 S \psi = S H_0 \psi \forall \psi \in \mathcal{D}(H_0) \cap \mathcal{D}(S)$.

Existence of the Møller Operators

Theorem (12.4 in [1]): Let $H = H_0 + V$ with $V \in L^2(\mathbb{R}^k)$ and $H_0 = -\Delta$. Then the $\Omega_{\pm}$ exists and $\mathcal{D}(\Omega_{\pm}) = \mathcal{H}$.

$\psi \in \mathcal{D}(\Omega_{\pm})$ is an eigenstate of H_0 if it is an eigenstate of H with the same eigenvalue.

4.1 Asymptotic Completeness

Asymptotic Completeness

Let $H = H_0 + V$ with $H_0 = -\Delta$. $\Omega_{\pm}$ (or V) is called **asymptotically complete** if

$$\mathcal{D}(\Omega_{\pm}) = \mathcal{H} \text{ and } \mathcal{H}_c(H) = \text{Ran}(\Omega_{\pm}).$$

If $\Omega_{\pm}$ is asymptotically complete, then $\mathcal{H}_{sc} = \{0\}$, S is **unitary** and kinetic energy is preserved during scattering, i.e. $\langle \psi_-, H_0 \psi_- \rangle = \langle \psi_+, H_0 \psi_+ \rangle$, where $\psi_+ := S \psi_-$.

B is said to be **relatively bounded** w.r.t. A if $\mathcal{D}(A) \subset \mathcal{D}(B)$ and if $\exists a, b \geq 0$ s.t. $\|B\psi\| \leq a\|A\psi\| + b\|\psi\| \forall \psi \in \mathcal{D}(A)$. The infimum of all such a for which a corresponding b exists such that the above holds is called the **A -bound** of B .

Short Range Potentials

Let V be relatively bounded w.r.t H_0 with bound < 1 . Let $\chi_r := \chi_{B_r^c}$. For $z \in \rho(H_0)$ and $r \geq 0$, define

$$h_{z,1}(r) := \|V(H_0 - z\mathbb{I})^{-1}\chi_r\|, \quad h_{z,2}(r) := \|\chi_r V(H_0 - z\mathbb{I})^{-1}\|.$$

Then V is called a **short range potential** if $h_{z,1}(r)$ and $h_{z,2}(r)$ are integrable. Indeed, $h_{z_0,i}$ is integrable for some $z_0 \in \rho(H_0)$ and some $i \in \{1, 2\}$ if and only if $h_{z,j}$ is integrable $\forall z \in \rho(H_0)$ and $\forall j \in \{1, 2\}$.

(For the proof of the following results, refer to section 12.3 of [1])

If **V is short range**, then:

- $(H - z\mathbb{I})^{-1} - (H_0 - z\mathbb{I})^{-1}$ is compact
- $\sigma_{ac}(H) = \sigma_{ess}(H) = [0, \infty)$, $\sigma_{sc}(H) = \emptyset$
- All nonzero eigenvalues of H have finite multiplicity and cannot accumulate in $(0, \infty)$.
- If $f : \mathbb{R} \rightarrow \mathbb{C}$ is continuous and vanish at ∞ , then $f(H) - f(H_0)$ is compact and $\lim_{r \rightarrow \infty} \|(f(H) - f(H_0))\chi_r\| = 0$.

RAGE theorem suggests that short range potentials should be asymptotically complete

Enß Theorem

Let V be a short range potential. Then the Møller operators are asymptotically complete.

4.2 Examples of Potentials

4.2.1 Coulomb potential

$V := \frac{\gamma}{|x|}$ is **not** a short range potential. It is known (Enß doesn't imply this) that the asymptotic states are not free[5].

4.2.2 Yukawa potential

$V := -g \frac{e^{-\mu|x|}}{|x|}$, $g, \mu > 0$ is a short range potential.

4.2.3 $\frac{C}{(1+|x|)^\alpha}$

If a potential satisfies $|V(x)| \leq \frac{C}{(1+|x|)^\alpha}$ for some $C > 0$, $\alpha > 1$, then V is a short range potential.

Bibliography

- [1] **Gerald Teschl**. Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators. 2nd ed., American Mathematical Society, 2014. <https://doi.org/10.1090/gsm/157>
- [2] **Ruelle, David**. A remark on bound states in potential-scattering theory. *Il Nuovo Cimento A* (1965-1970) 61 (1969): 655-662. <https://api.semanticscholar.org/CorpusID:56050354>
- [3] **Amrein, W. O. and V. Georgescu**. On the Characterization of Bound States and Scattering States in Quantum Mechanics. *Helv. Phys. Acta*, vol. 46, no. 5, Jan. 1974, pp. 635-58. <https://inis.iaea.org/records/5hk4c-rrx72>
- [4] **Enss, V**. Asymptotic completeness for quantum mechanical potential scattering. *Commun.Math. Phys.* 61, 285-291 (1978). <https://doi.org/10.1007/BF01940771>
- [5] **John D. Dollard**. Asymptotic Convergence and the Coulomb Interaction. *J. Math. Phys.* 1 June 1964; 5 (6): 729-738. <https://doi.org/10.1063/1.1704171>